

MADHURA JOSHI

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EDUCATION

B.Tech, Chemical Engineering | Indian Institute of Technology (IIT), Madras

CGPA: 7.87/10 | 2024 - 2028*

- Academic achievements:** Secured **99.63 percentile** in JEE-Mains 2024 (AIR 5950); Placed in the top tier of JEE Advanced (AIR 9060) out of 1.5L+ candidates.
- Relevant Coursework:** Programming and Data Structures in Python, Machine Learning Foundations, Computational Techniques, Probability & Statistics, Kinetics & Catalysis, Chemical Engineering Thermodynamics.

TECHNICAL SKILLS

Programming Languages:	Python (Expert), SQL, MATLAB, JavaScript, Java
Libraries & Frameworks:	PySpark, Scikit-Learn, Pandas, NumPy, Plotly, PyTorch, Matplotlib, Seaborn
Tools & Enterprise Systems:	Streamlit, SQLite (LIMS Simulator), Git, GitHub, Microsoft Excel, Power BI
Core Competencies:	Formulation Science Modeling, Property Simulation, AI Agent Workflows, Process Mapping, Multi-Output Regression

PROFESSIONAL EXPERIENCE

Machine Learning Intern | [Company / Research Lab]

May 2025 - June 2025

- R&D; Data Pipelines:** Designed Python data pipelines to clean, preprocess, and structure experimental measurements, streamlining exploratory data analysis (EDA) for new formulations.
- Predictive Modeling:** Trained and evaluated predictive machine learning models in Scikit-Learn to reduce physical lab trial-and-error cycles.
- Data Storytelling:** Visualized multi-variable experimental correlations and key physical trends, communicating findings to senior R&D; team leads.

Co-Founder & Business Lead | #DotConnect (DotEco Founding Circle)

Dec 2025 - Present

- Process Mapping:** Mapped business workflows and established key performance indicators (KPIs) to track user growth metrics.
- Visual Analytics:** Built interactive business performance tracking dashboards, supporting executive decision-making.

TECHNICAL PROJECTS

FormuAI: Formulation Optimizer & Digital Lab Agent (FMCG R&D;)

June 2026

- Formulation Modeling:** Trained a multi-output RandomForestRegressor model on synthetic ingredient ratios (surfactants, co-surfactants, thickeners, actives) to predict critical shampoo properties (viscosity, pH, foam volume, stability), reducing trial-and-error cycles from days to < 10ms.
- AI Lab Copilot Agent:** Developed an NLP search and optimization agent that translates chemist instructions (e.g. 'Show experiments with stability > 300 days') into SQLite SQL commands or simulation parameters.
- LIMS & BI Integration:** Set up a relational SQLite database simulating LIMS/ELN records and built an interactive Streamlit dashboard mapping ingredient correlations, mimicking enterprise Power BI reports.

LumiFoam: Computer Vision Foam & Emulsion Analyzer

May 2026

- Computer Vision Pipeline:** Built OpenCV image processing pipelines using Hough Circle Transforms to automate droplet size measurements of surfactant emulsions.
- Decay Curve Fitting:** Applied SciPy exponential curve-fitting models to calculate shampoo foam volume decay half-life, automating physical lab measurements.

Adsorption Modeling in Porous Solids & Heterogeneous Systems

May 2026

- Numerical Solvers:** Formulated a MATLAB mathematical model solving coupled partial differential equations (PDEs) for mass and energy transport in heterogeneous catalyst pellets.

- **Optimization:** Analyzed internal reactant distributions and Thiele modulus across complex geometries to prevent starvation regimes.

POSITIONS OF RESPONSIBILITY

Club Head - Fine Arts Sangam, IIT Madras	May 2025 - June 2026
• Cross-Functional Leadership: Governed financial budgeting and led a multi-tiered team of 9 coordinators and a 35-member contingent. • Process & Documentation: Drafted operational blueprint reports and financial decks, coordinating event marketing campaigns that increased engagement by 20%.	